

“We can only see a short distance ahead, but we can see **plenty** there that **needs to be done.**”

Alan Turing, 1950

Turing said of his machine: “We have only to regard the rules as being capable of being taken out and exchanged for others and we have something very akin to a universal computing machine.” Although the technology did not yet exist to build one, the “Turing machine” hinted at the limitless potential of the future field of computer science.

Code-breaker

After 2 years in the US at Princeton University—where he gained his PhD—Turing was approached by the

British government's Code and Cypher School in 1938. The school had been set up to decipher the German military's coded radio messages—the trouble was, the Germans were using an encryption device known as Enigma, which scrambled their messages in different ways each day. Indeed, the Germans even boasted that the Enigma code was unbreakable. With this daunting task ahead of him, Turing left Cambridge—where he had begun building a mechanical computing device—and joined the Cypher School at Bletchley Park. He arrived on September

